

EXPERIMENT 20: PERFORM SHARPENING OF IMAGE USING LAPLACIAN MASK WITH NEGATIVE CENTER COEFFICIENT

Aim

To sharpen an image using a Laplacian mask with a negative center coefficient in Python and OpenCV.

Algorithm

1. Import OpenCV and NumPy libraries.
2. Read the input image.
3. Convert the image to grayscale.
4. Define the Laplacian kernel with a negative center coefficient.
5. Apply the Laplacian filter using filter2D().
6. Subtract the Laplacian image from the original image to obtain a sharpened image.
7. Display and save the output image.

Python Code

```
import cv2
import numpy as np

img = cv2.imread("input.jpg")

if img is None:
    print("Image not found")
    exit()

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

kernel = np.array([[0, 1, 0],
                   [1,-4, 1],
                   [0, 1, 0]], dtype=np.float32)

laplacian = cv2.filter2D(gray, -1, kernel)
```

```
sharpened = cv2.subtract(gray, laplacian)

cv2.imwrite("laplacian_output.jpg", sharpened)

cv2.imshow("Original Image", gray)
cv2.imshow("Sharpened Image", sharpened)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input

0	1	0
1	-4	1
0	1	0

Output

0	1	0
1	-4	1
0	1	0

Result

Thus, the image was successfully sharpened using the Laplacian mask with a negative center coefficient.